



MINHAZUL ISLAM

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[in MINHAZUL ISLAM](#) | [Portfolio](#)

Ningbo, Zhejiang | 愿意搬迁

SUMMARY

AI/ML 工程师, 拥有硕士学位, 深耕算法设计、生成式 AI (Generative AI) 及自然语言处理 (NLP) 系统。具备端到端机器学习工程的丰富经验, 精通大语言模型 (LLM) 微调、RAG 检索增强生成流水线, 以及基于 LangGraph、LangChain、n&n 和 Dify 的多智能体编排 (Multi-agent Orchestration)。熟练掌握 FastAPI 构建高并发推理服务, 能够部署可扩展的生产级 AI 系统。具备将复杂算法转化为实际应用的能力, 并在系统部署、性能优化及同行评审研究 (Peer-reviewed Research) 方面有坚实成果。

EDUCATION

- 浙江大学 2023 年 9 月至今
工业设计工程硕士 中国浙江省
- 云南大学 2019-2023
计算机科学与技术学士 中国昆明

项目经历

- AI 驱动的双相情感障碍同伴支持平台 2025
浙江大学 [Link](#)
 - 核心技术栈: Python, Flask, FastAPI, PyTorch, RoBERTa, LangChain, DeepSeek LLM, Sentence Transformers, ChromaDB, RAG, Pushshift API, ThreadPoolExecutor
 - 开发了一个功能完善的网络版智能写作助手原型, 旨在为同伴支持者提供实时反馈, 评估其回复的情感支持 (ES) 与信息支持 (IS) 水平, 并利用生成式人工智能提出情境感知的改进建议, 以生成更安全、更具共情力的回复。
 - 设计并部署了可扩展的微服务架构, 集成了大语言模型 (LLM)、检索增强生成 (RAG) 以及基于 Transformer 的回归模型。
 - 构建了稳健的数据管道, 用于爬取并预处理过去六年 Reddit 上的心理健康讨论数据。带领三名领域专家进行了严谨的标注工作, 标注信息支持与情感支持 (IS/ES) 指标, 创建了一个用于监督学习的高质量黄金标准数据集。
 - 在专家标注的数据集上对 RoBERTa 模型进行微调, 以预测与人类判断高度相关的支持质量分数, 通过迁移学习使预训练的 Transformer 模型适应心理健康支持领域。
 - 使用 LangChain、ChromaDB 和 SentenceTransformers 构建了检索增强生成 (RAG) 系统, 检索语义相似、高质量的人工回复作为 DeepSeek LLM 的生成基础, 从而减少幻觉并提升生成建议的质量。
 - 通过高并发 FastAPI 推理服务部署模型, 并与 Flask 前端解耦; 通过异步请求处理以及并行执行 LLM 生成与安全检查 (使用 ThreadPoolExecutor), 将推理延迟优化了约%。
 - 开发了基于 LLM 的实时安全过滤中间件, 用于检测并阻止有害内容、自残风险及危险的医疗建议, 确保严格遵守心理健康应用的伦理准则。
 - 实现了一个自动化的主动学习管道, 能够过滤、伪标注并将新的用户提交内容纳入训练集, 使模型能够在无需人工干预的情况下自适应并持续改进。

核心工程成果:

- 全栈 AI 部署: 成功将复杂的深度学习模型 (RoBERTa, LLMs) 集成至用户友好的 Web 应用程序, 打通了研究代码与生产软件。
- 可扩展的向量搜索: 优化 ChromaDB 索引, 实现了对数千份文档的亚毫秒级检索。

- 以用户为中心的验证：通过对 24 名双相情感障碍患者开展效果研究，验证了该助手能有效降低认知负荷并提升同伴支持质量。

SPROUT——面向学业心态支持的形变冥想装置

2024

浙江大学

关联 

- **技术栈：**Arduino IDE/C++、嵌入式系统、传感器融合、Fusion 360
- 设计生物启发的形变交互板，帮助学生将学业挑战转化为成长契机。
- 针对“失败恐惧”心理压力，开发非侵入式支持干预装置。
- 基于 Arduino 集成传感器与执行器，打造冥想式交互，引导用户反思心态。
- 采用“播种”隐喻，将负面情绪转化为成长动力，增强心理韧性。
- 技能：系统设计、嵌入式开发、电路原型、C++ 编程、传感器集成、原型迭代。

AgenticAI 旅行规划师——基于大型语言模型的智能度假助手

2025

个人项目

Link 

- **工具：**Python、Google 智能体开发工具包、大型语言模型（Gemini 2.5）、模块化智能体架构
- 开发以用户为中心的旅行规划助手，运用大型语言模型与智能体设计实现个性化、可操作的行程推荐。
- 构建包含主智能体与子智能体的模块化系统，根据用户偏好与限制收集、整合并呈现定制化旅行洞察。
- 集成外部 API 与工具，实时获取目的地、活动及兴趣点数据，提升推荐质量。
- 通过自动化调研、方案摘要及下一步行动/替代方案提示，打造清晰流畅的用户体验。
- 应用技能：大型语言模型集成、智能体系统设计、API 集成、用户体验设计、Python 编程。

Autodesk Inventor API + Python——参数化 CAD 自动化

2024

ProFabx

关联 

- **技术栈：**Autodesk Inventor、Python、win32com.client
- 利用 Python 构建参数化建模系统，通过修改关键尺寸动态更新 Inventor 3D 模型。
- 自动化重复性 CAD 任务，缩短设计周期并减少人工错误。
- 将 Python 脚本深度集成至 Inventor API，优化机械设计流程，提升原型迭代效率。

InclusiveVision——视障人士智能眼镜

2023

浙江大学

关联 

- **技术栈：**Arduino、Fusion 360、Flashforge 3D 打印、Talkie 库、激光切割
- 基于 Arduino UNO 与超声波传感器开发智能眼镜原型，实现 50 cm 内实时障碍物检测。
- 使用 Talkie 语音库构建警报系统，实时处理传感器数据并提供即时听觉反馈。
- 在 Fusion 360 中完成眼镜框架 3D 建模，兼顾可穿戴舒适度与功能性。
- 设计集成多颗超声波传感器的模块化组件，便于快速装配，全面提升视障用户独立性与安全性。

基于协同过滤算法的产品推荐系统

2023

云南大学

关联 

- **技术栈：**Python、NumPy、Pandas、MongoDB、Matplotlib、Flask
- 针对 goodbooks-10k 数据集开发协同过滤推荐策略，生成高质量书单。
- 利用 Python 及 NumPy、Pandas、Matplotlib 完成数据分析与可视化，确保推荐精准度。
- 基于矩阵分解构建推荐模型，为用户提供个性化图书建议。
- 引入欧氏距离度量书籍相似度，进一步提升推荐准确率。

基于 IoT 与人脸识别的智能考勤系统

2022

云南大学

关联 

- **技术栈：**ESP32-CAM、Arduino IDE/C++、嵌入式系统、HTTPS、IoT 云
- 采用 ESP32-CAM 开发 IoT 智能考勤系统，通过人脸识别自动完成点名。
- 实现安全认证流程：系统自动检测、识别并注册学生，通过 HTTPS 将数据上传至 IoT 云端。

- 针对新冠疫情设计零接触方案，替代传统人工及接触式 biometric 系统。
- 仅验证已注册用户，杜绝代打卡，提升考勤可靠性与安全性。
- 综合运用嵌入式系统、电路原型与 C++ 编程，完成软硬件一体化集成。

EXPERIENCE

- **ProFabx** June 2024 - September 2024
Software Engineer Ningbo, China

发表论文

- [C.1] Minhazul Islam, Mengru Xue, and Tasnim Afra (2025). **Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework**. In *Proceedings of the 24th IFIP International Conference on Entertainment Computing (ICEC 2025)*, Tokyo, Japan. doi:10.1007/978-3-032-02534-0₈
- [C.2] Tasnim Afra, Mengru Xue, and Minhazul Islam (2025). **Enhancing Biofeedback Interventions for Depression and Anxiety through Entertainment Computing: A Systematic Review**. In *Proceedings of the 24th IFIP International Conference on Entertainment Computing (ICEC 2025)*, Tokyo, Japan. doi:10.1007/978-3-032-02534-0₆

认证与技术培训

- **使用大型语言模型的生成式人工智能** | DeepLearning.AI | 2025 年 11 月 | [查看证书](#)
掌握了大型语言模型的生命周期，涵盖了扩展规律、计算挑战和模型评估指标。应用了 PEFT 和 LoRA 进行模型自适应，并实现了 RLHF 进行人机对齐。
- **精通生成式人工智能：Transformer 微调** | IBM | 2025 年 10 月 | [查看证书](#)
针对特定分类和命名实体识别 (NER) 任务，对 BERT 和 RoBERTa 架构进行微调。使用量化 (QLoRA) 和针对特定领域文本的自定义分词策略优化模型。
- **使用 Python 开发生成式 AI 应用** | IBM | 2025 年 10 月 | [查看证书](#)
使用 Flask 构建应用程序，并集成 IBM Watson 和 OpenAI API。使用 Docker 和 Gradio 构建容器化的语音助手和 ChatGPT 风格的界面。
- **面向人工智能和开发的 Python 项目** | IBM | 2025 年 9 月 | [查看证书](#)
应用了包括单元测试 (Pytest)、静态代码分析和 Python 打包在内的专业实践。使用 NumPy 和 Pandas 执行了数据清洗和特征工程。

专业技能

- **生成式人工智能与自然语言处理**: 大型语言模型 (GPT-3、DeepSeek)、RAG、Dify (大型语言模型应用套件)、n8n (工作流程自动化)、LangGraph、LangChain、LlamaIndex、Transformers (BERT, RoBERTa)、Hugging Face、命名实体识别 (NER)。
- **机器学习与运维**: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, 模型评估, **Docker, Kubernetes**, CI/CD, Git, 单元测试 (Pytest)。
- **数据与向量数据库**: ChromaDB, Pinecone, Elasticsearch, MongoDB, PostgreSQL, ETL 流水线 (Apache Airflow), NumPy, Pandas, Matplotlib, Seaborn。
- **语言与性能优化**: **Python (Asyncio)**, C/C++, JavaScript, ThreadPoolExecutor (多线程并发), API 集成 (REST), 静态代码分析。
- **全栈开发与部署**: **FastAPI (高并发推理)**, Flask, Node.js, React.js, Tailwind CSS, D3.js, HTML5/CSS3, IBM Watson 集成。
- **嵌入式与自动化**: 边缘 AI (Edge AI), 物联网 (ESP32, Arduino), 传感器融合, Fusion 360, Autodesk Inventor API, 参数化设计, Python-CAD 集成。



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MINHAZUL ISLAM | Portfolio

Ningbo, Zhejiang | Open to relocate

SUMMARY

AI/ML Engineer with a Master's degree, specializing in algorithm design, Generative AI, and NLP systems. Strong experience in end-to-end machine learning engineering, including LLM fine-tuning, RAG pipelines, and multi-agent orchestration using LangGraph, LangChain, and dify, as well as high-concurrency inference services (FastAPI). Proven ability to deploy scalable, production-ready AI systems and translate algorithms into real-world applications, supported by system deployment, performance optimization, and peer-reviewed research.

EDUCATION

- | | |
|--|---|
| • Zhejiang University
<i>Masters in Industrial Design Engineering</i> | September 2023 - Present
Zhejiang, China |
| • Yunnan University
<i>Bachelor in Computer Science and Technology</i> | 2019-2023
Kunming, China |

PROJECTS

- **AI-Powered Mental Health Peer Support Platform** 2025
Zhejiang University Link[🔗]
 - **Tools:** Python, Flask, FastAPI, PyTorch, RoBERTa, LangChain, DeepSeek LLM, Sentence Transformers, ChromaDB, RAG, Pushshift API, ThreadPoolExecutor
 - Developed a fully functional web-based prototype as an intelligent writing assistant for peer supporters, providing real-time feedback on Emotional Support (ES) and Informational Support (IS) levels and using Generative AI to suggest context-aware improvements for safer, more empathetic replies.
 - Designed and deployed a scalable microservice architecture, integrating a large language model (LLM), retrieval augmentation generation (RAG), and a Transformer-based regression model.
 - Built a robust data pipeline to crawl and preprocess mental health discussions from Reddit over the past six years. Led a rigorous annotation effort with three domain experts to annotate informational and emotional support (IS/ES) metrics, creating a high-quality gold-standard dataset for supervised learning.
 - Experimented with various machine learning models—SVM, Multinomial Logistic Regression, Random Forest (RF), and XGBoost—to predict support metrics, achieving the best results for IS with RF and the best results for ES with XGBoost.
 - Fine-tuned the RoBERTa model on the expert-annotated dataset to predict support quality scores highly correlated with human judgment.
 - Built a retrieval augmentation generation (RAG) system using LangChain, ChromaDB, and SentenceTransformers, retrieving semantically similar, high-quality human responses to ground the DeepSeek LLM, reducing hallucinations and improving generated suggestions.
 - Deployed models via a high-concurrency FastAPI inference service, decoupled from the Flask frontend, optimizing inference latency by approximately 65% through asynchronous request processing and parallel execution of LLM generation and security checks (using ThreadPoolExecutor).
 - Developed a real-time security filtering middleware using LLM to detect and block harmful content, self-harm risks, and dangerous medical suggestions, ensuring strict adherence to ethical guidelines for mental health applications.

- Implemented an automated active learning pipeline that filters, pseudo-labels, and incorporates new user submissions into the training set, enabling the model to adapt and improve over time without human intervention.

SPROUT –Shape-changing Meditation Device for Academic Mindset Support

2024

Zhejiang University

[Link](#)

• Tools: Arduino IDE/C++, Embedded Systems, Sensor Integration, Fusion 360

- Designed a bio-inspired, shape-changing interactive board to help students turn academic challenges into opportunities for growth.
- Targeted the psychological stressor “Fear of Failure” by developing a supportive, non-intrusive intervention device.
- Integrated sensors and actuators with embedded systems to create a meditative interaction, helping users reflect on their mindset.
- Conducted mixed-methods user studies with university students, combining stress and mood assessments with qualitative interviews on usability and emotional engagement.
- Applied human-centered design, embodied interaction, and HCI evaluation methods to create an intuitive, multisensory interface. Demonstrated how tangible bio-inspired interfaces can enhance relaxation, meditation, and personal growth.
- Skills applied: System Design, Embedded Systems, Circuit Prototyping, C++ Programming, Sensor Integration, Prototype Development.

AgenticAI Travel Planner –LLM-based Agentic Vacation Assistant

2025

Personal Project

[Link](#)

• Tools: Python, Google Agent Development Kit, LLMs (Gemini 2.5), Modular Agentic Architecture

- Developed a user-centric travel planning assistant leveraging large language models and agentic design for personalized, actionable trip recommendations.
- Architected a modular system with a main agent and sub-agents to gather, synthesize, and present travel insights tailored to user preferences and constraints.
- Integrated external APIs and tools for real-time destination, event, and point-of-interest data, enhancing recommendation quality.
- Focused on clear, engaging user experience by automating research, summarizing options, and providing next steps or alternatives.
- Skills applied: LLM Integration, Agentic System Design, API Integration, User Experience, Python Programming.

Autodesk Inventor API with Python –Parametric CAD Automation

2024

ProFabx

[Link](#)

• Tools: Autodesk Inventor, Python, win32com.client

- Created a parametric modeling system using Python to dynamically update Autodesk Inventor 3D models by modifying key dimensions.
- Automated repetitive CAD tasks, reducing design time and minimizing manual errors.
- Integrated Python scripting with Inventor’s API to streamline mechanical design workflows and improve prototyping efficiency.

InclusiveVision - Smart Glasses for Visually Impaired Individuals

2023

Zhejiang University

[Link](#)

• Tools: Arduino, Fusion 360, Flashforge 3D Printing, Talkie Library, Laser Cutting

- Developed a prototype of smart glasses using an Arduino UNO and ultrasonic sensors, achieving real-time obstacle detection up to 50 cm.

- Implemented a voice alert system using the Talkie library, processing sensor data to provide immediate auditory feedback to users.
- Created 3D models of the glasses frame with Fusion 360, optimizing the design for comfort and functionality in a wearable form factor.
- Developed an integrated component with multiple ultrasonic sensors for easy assembly, enhancing the independence and safety of visually impaired users.

Product recommendation system based on Collaborative Filtering Algorithm

2023

Yunnan University

Link[🔗]

- **Tools:** Python, NumPy, Pandas, MongoDB, Matplotlib, Flask
- Developed a collaborative filtering strategy for the goodbooks-10k data-set to generate high-quality recommendations.
- Implemented Python and related libraries (NumPy, Pandas, Matplotlib) for data analysis and visualization to achieve accurate recommendations.
- Created a recommendation model based on matrix factorization, ensuring personalized suggestions for users.
- Applied Euclidean Distance to analyze the similarity between books, enhancing the accuracy of recommendations.

IoT-based Smart Attendance System with Face Recognition

2022

Yunnan University

Link[🔗]

- **Tools:** ESP32-CAM, Arduino IDE/C++, Embedded Systems, HTTPS, IoT Cloud
- Developed a smart, IoT-based attendance system using an ESP32-CAM for facial recognition to automate the attendance-taking process.
- Implemented a secure authentication workflow where the system automatically detects, recognizes, and registers students, storing data in an IoT cloud via HTTPS.
- Designed a contact-free solution to address safety concerns from the COVID-19 pandemic, replacing traditional manual and contact-based biometric systems.
- Created a system to prevent proxy attendance by only validating registered users, enhancing the reliability and security of the process.
- Applied skills in embedded systems, circuit prototyping, and C++ programming to integrate the hardware and software components.

EXPERIENCE

- ProFabx

Software Engineer

June 2024 - September 2024

Ningbo, China

PUBLICATIONS

- [C.1]

Minhazul Islam, Mengru Xue, and Tasnim Afra (2025). **Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework.** In *Proceedings of the 24th IFIP International Conference on Entertainment Computing (ICEC 2025)*, Tokyo, Japan. [doi:10.1007/978-3-032-02534-0_8](https://doi.org/10.1007/978-3-032-02534-0_8)
- [C.2]

Tasnim Afra, Mengru Xue, and Minhazul Islam (2025). **Enhancing Biofeedback Interventions for Depression and Anxiety through Entertainment Computing: A Systematic Review.** In *Proceedings of the 24th IFIP International Conference on Entertainment Computing (ICEC 2025)*, Tokyo, Japan. [doi:10.1007/978-3-032-02534-0_6](https://doi.org/10.1007/978-3-032-02534-0_6)

CERTIFICATIONS & TECHNICAL TRAINING

- **Generative AI with Large Language Models** | DeepLearning.AI | Nov 2025 | [View Certificate](#)
Mastered the LLM lifecycle, covering scaling laws, computational challenges, and model evaluation metrics. Applied PEFT and LoRA for model adaptation and implemented RLHF for human-alignment.
- **Mastering Generative AI: Fine-Tuning Transformers** | IBM | Oct 2025 | [View Certificate](#)
Fine-tuned BERT and RoBERTa architectures for specialized classification and NER tasks. Optimized models using Quantization (QLoRA) and custom tokenization strategies for domain-specific text.
- **Developing Generative AI Applications with Python** | IBM | Oct 2025 | [View Certificate](#)
Architected applications using Flask, integrating IBM Watson and OpenAI APIs. Built containerized Voice Assistants and ChatGPT-style interfaces using Docker and Gradio.
- **Python for AI & Development Project** | IBM | Sep 2025 | [View Certificate](#)
Applied professional practices including Unit Testing (Pytest), Static Code Analysis, and Python packaging. Executed data cleaning and feature engineering using NumPy and Pandas.

SKILLS

- **Generative AI & NLP:** LLMs (GPT-3, DeepSeek), RAG, Dify (LLM App Stack), n8n (Workflow Automation), LangGraph, LangChain, LlamaIndex, Transformers (BERT, RoBERTa), Fine-tuning (PEFT, LoRA, QLoRA), Prompt Engineering, Hugging Face, Gradio, NER
- **Machine Learning & Ops:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest (RF), SVM, Logistic Regression, Model Quantization, Model Evaluation, Docker, Kubernetes, CI/CD, Git, Unit Testing (Pytest)
- **Data Science & Databases:** Elasticsearch, MongoDB, ChromaDB, Pinecone, PostgreSQL, NumPy, Pandas, Matplotlib, Seaborn, ETL Pipelines (Apache Airflow), Data Loaders, Feature Engineering.
- **Languages & Performance:** Python (Asyncio), C/C++, JavaScript, API Integration (REST), ThreadPoolExecutor (Concurrency), Win32com, Static Code Analysis.
- **Full-Stack Development:** FastAPI, Flask (Web App Deployment), Node.js, React.js, Tailwind CSS, D3.js, HTML5/CSS3, IBM Watson Integration.
- **Embedded & Automation:** IoT (ESP32, Arduino), Edge AI, Sensor Fusion, Fusion 360, Autodesk Inventor API, Parametric Design, Python-CAD Integration.